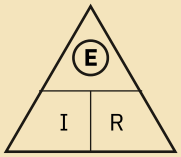
Cover the circled letter

Side by side means multiply. One over the other means divide. E volts, I amps, R ohms, P watts.

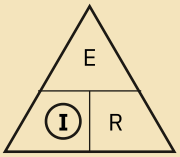
OHM'S LAW

Voltage



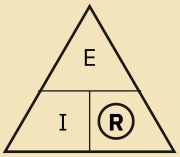
E = I × R

Current



I = E / R

Resistance

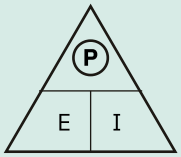


R = E / I

Book example: 90 volts and 3 amps. Cover R. 90 / 3 = **30 ohms**. 12 volts across 4 ohms is 3 amps.

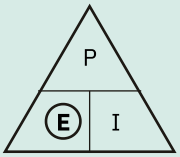
POWER

Power



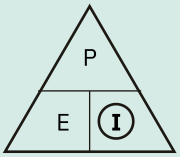
P = E × I

Voltage



E = P / I

Current



I = P / E

13.8 V × 10 A = **138 W**. 12 V × 2.5 A = 30 W. 120 W at 12 V is 10 A.

POWER WHEEL · I SQUARED

P = I² × R
I = √(P / R)
R = P / I²

If the cell says I², divide first, then take the square root to get amps.

POWER WHEEL · E SQUARED

E = √(P × R)
P = E² / R
R = E² / P

E² is on top because E² = P × R. Square root that to get volts.

PREFIXES · EACH STEP IS ×1,000

pico, nano, micro, milli, the unit, kilo, mega, giga.
1.5 A = 1,500 mA. 28,400 kHz = 28.4 MHz. 500 mW = 0.5 W. Write MHz and kHz, not just "mega."
Decibels. Twice the power is about +3 dB. Ten times is +10 dB. Down to a quarter is about −6 dB.

Electrical ideas

Voltage is push. Current is flow. Resistance is the squeeze. Power is push times flow.

THE NAMES

- **E** voltage, volts. Also called EMF.
- **I** current, amperes. Electrons moving.
- **R** resistance, ohms. Opposes current.
- **P** power, watts. The work being done.
- **Hz** frequency, cycles each second.
- **F** capacitance, farads. Stores energy in an electric field.
- **H** inductance, henries. Stores energy in a magnetic field.
- **Z** impedance, ohms. The squeeze against AC.

CONDUCTORS AND INSULATORS

Conductors let current flow: copper, aluminum, gold, silver, most metals, salt water.
Insulators block it: glass, rubber, ceramic, plastic, dry air, wood.
A wire is a conductor wrapped in an insulator. A short is almost no resistance. An open is a broken path, so current stops.

DC AND AC

DC current flows one way. A battery is DC. The trace is a flat line.
AC current reverses over and over. A wall outlet and a radio wave are AC. The trace is a wave.
Frequency is how many full cycles happen in one second. 60 Hz reverses 60 times a second. A 146 MHz signal does it 146 million times a second.

SERIES AND PARALLEL

Series: one path. Current is the same everywhere. Voltages add. Resistances add.
Parallel: branches. Voltage is the same across each branch. Currents add. Total resistance is less than the smallest resistor.
Two equal resistors in parallel equal half of one of them.

HOW YOU MEASURE

A **voltmeter** goes across the part, in parallel. An **ammeter** goes in the path, in series. An **ohmmeter** is used with the power off. A multimeter does all three. Never measure ohms on a live circuit.

THREE MORE WORDS THAT SHOW UP TOGETHER

A **capacitor** passes AC and blocks DC. An **inductor** (a coil) passes DC and fights a change in current. **Impedance** is the total opposition to AC: resistance plus the effect of capacitors and coils. It is still measured in ohms.

Rules and where you may transmit

The FCC makes the rules. A band plan is only a gentleman's agreement. If they disagree, follow the FCC.

YOUR LICENSE

Element 2 is the Technician test: 35 questions, 26 to pass. Three Volunteer Examiners give the test. The license lasts **10 years**. You have a 2-year grace period to renew, but you may not transmit until it is renewed. You may operate when your call sign appears in the FCC database, not before. The control operator's license sets the privileges. The station owner is still responsible. Say your call sign every **10 minutes** and at the end of the contact.

Band	Frequencies	What a Technician may do
80 m	3.525–3.600 MHz	CW only, 200 watts
40 m	7.025–7.125 MHz	CW only, 200 watts
15 m	21.025–21.200 MHz	CW only, 200 watts
10 m	28.000–28.300 MHz	CW and data, 200 watts
10 m	28.300–28.500 MHz	Phone and CW, 200 watts. Your only HF voice.
6 m	50–54 MHz	CW only 50.0–50.1. All modes above that. Up to 1500 W.
2 m	144–148 MHz	All modes, up to 1500 W
1.25 m	222–225 MHz	All modes, up to 1500 W
70 cm	420–450 MHz	All modes. Amateurs are secondary here.
33 cm / 23 cm	902–928 and 1240–1300 MHz	All modes

NOT YOUR BANDS

20, 17, 12, and 60 meters are not Technician bands. Neither is most of 80, 40, 15, and 10. Above 30 MHz the power cap is 1500 watts, except a few special slices. Always use the **least power that makes the contact**.

DO NOT TRANSMIT

- Music, broadcasting, obscenity, or false calls.
- Hidden messages. Codes are allowed only to control a satellite.
- Anything for your job or for pay.
- When you would interfere with a primary user. Stop.

PEOPLE ON THE AIR

The **control operator** is the licensed ham responsible at the control point. A non-ham may speak only as third-party traffic, with you there to ID. Third-party messages to other countries are allowed only where the US has an agreement. One-way transmissions are allowed for beacons, code practice, emergency traffic, and bulletins aimed at other hams. That is not broadcasting.

Operating

Listen first. Leave a pause so someone else can break in. Say your call sign.

REPEATER

A repeater listens on one frequency and transmits on another, from a high place, so handhelds can reach each other.
You listen on the output. You transmit on the input. The split is the **offset**: usually **600 kHz** on 2 meters and **5 MHz** on 70 cm.
A **CTCSS** tone (you usually cannot hear it) opens the repeater. A courtesy tone means the machine has dropped, so you may talk. Don't "kerchunk" just to hear it.

SIMPLEX

Simplex means both people use the same frequency. Use it when you can hear each other without the repeater, so the repeater stays free.
146.520 MHz is the 2 meter FM calling frequency.
446.000 MHz is the 70 cm FM calling frequency.
144.390 MHz is the US APRS frequency.

A CONTACT

- Calling anyone: "CQ". On a repeater, just say your call sign.
- Use phonetics. Alpha, Bravo, Charlie…
- On a directed net, talk when net control asks.
- Emergency: stop, listen, help only if you can. Say "emergency."
- RST report for Morse: readability 1–5, strength 1–9, tone 1–9. A good signal is 599.

Q SIGNALS

- **QRM** man-made interference
- **QRN** static, natural noise
- **QSY** change frequency
- **QTH** location
- **QRZ** who is calling me
- **QSB** fading
- **QSL** I confirm

WHICH BAND IS THAT FREQUENCY?

3.5 MHz is 80 m. 7 MHz is 40 m. 21 MHz is 15 m. 28 MHz is 10 m. 52 MHz is 6 m. 146 MHz is 2 m. 223 MHz is 1.25 m. 443 or 446 MHz is 70 cm. 1296 MHz is 23 cm. Wavelength in meters is about **`300 / MHz`**. A 146 MHz signal is about 2 meters long.

How a radio signal travels

Radio waves move at the speed of light. Wavelength in meters = 300 ÷ frequency in MHz.

HF CAN SKIP

HF is 3–30 MHz. Those waves can bounce off the ionosphere and reach other countries. More sunspots, better HF, especially 10 and 15 meters.
The **F layer** is the mirror for long-distance skip. The **D layer** appears in the daytime and soaks up lower HF, so 80 meters is better at night.

VHF AND UHF ARE LOCAL

Above 30 MHz the ionosphere usually lets the wave go out into space. Range is a bit past the visual horizon, because the air bends the wave slightly.
Buildings and leaves soak up UHF more than VHF. A handheld inside a car or a building is weaker.

SPORADIC E

A sudden patch in the E layer. Surprise long contacts on 6 meters and 10 meters. 6 meters is called the magic band for this.

TROPO DUCTING

A weather layer bends VHF and UHF and carries them hundreds of miles. Common over water and in a temperature inversion.

KNIFE-EDGE

A sharp ridge bends some of the wave down into the valley on the far side. That is how a mountain does not always block you.

PICKET FENCING

A mobile signal flutters. The wave arrived by more than one path, and the copies add and cancel as you drive. That is multipath.

POLARIZATION

Hold the antenna the same way as the station you want. Most FM and repeaters are vertical. A horizontal dipole is horizontal. A mismatch can cost a lot of signal.

EXAMPLES

300 / 146 ≈ 2 m, so 146 MHz is the 2 meter band. 300 / 50 = 6 m. 300 / 440 ≈ 0.7 m, which is 70 cm. A longer wavelength is a lower frequency.

The station

A bad match sends power back down the cable. Fix the antenna. Don't just turn the radio up.

SWR

1:1 means the antenna and the cable agree. Power goes out. A high SWR means some power bounces back. Causes: wrong length, broken wire, bad connector, antenna against metal, water in the coax.
Under about 2:1 is usually fine. A very high SWR can damage the radio. Many radios fold back the power to protect themselves.

TUNER AND DUMMY LOAD

An **antenna tuner** makes the radio see a match. It does not make a poor antenna radiate better. It protects the radio.
A **dummy load** is a shielded resistor that acts like an antenna and turns the power into heat. Use it to test the radio without putting a signal on the air.

FEED LINE

Radios want about **50 ohm** coax. Loss grows with frequency and with length. At UHF, use a short run of good coax. Seal outdoor connectors so water cannot wick in. Ladder line is low loss but must stay clear of metal.

METERS

Voltmeter in parallel. Ammeter in series. Ohmmeter with power off and the part out of the circuit if you can. An S meter shows relative received strength, not watts.

RF IN THE SHACK, AND EXPOSURE

A ferrite choke on the coax stops RF from coming back into the mic and the computer. A low-pass filter after the transmitter cuts harmonics. Exposure depends on power, duty cycle, frequency, antenna gain, and distance. FM and digital are high duty cycle because the carrier stays on. SSB is lower. If you are over the limit: lower the power, raise or move the antenna, or transmit less.

Parts and the symbols

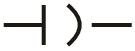
The test shows a symbol and asks its name. Learn the pictures, not just the words.



Resistor



Capacitor



Coil / inductor



Diode · bar is the cathode



LED · light comes out



NPN transistor · arrow out



Ground



Antenna

WHAT THEY DO

- **Resistor** sets current. A potentiometer is an adjustable one.
- **Capacitor** stores energy in an electric field. Blocks DC, passes AC.
- **Inductor** stores energy in a magnetic field. A coil.
- **Diode** lets current one way. The bar is the cathode.
- **LED** a diode that lights. Still one way.
- **Transistor** a switch or an amplifier. NPN arrow points out.
- **Transformer** two coils. Changes voltage. No DC path between sides.
- **Fuse** opens when current is too high. Put it in the hot lead.
- **Relay** a switch pulled in by a coil.
- **Battery** a stack of cells. Long line is positive.

READING A SCHEMATIC

Lines are wires. A dot is a connection. A jump with no dot is not a connection. Ground symbols are the common return, not always the dirt outside.

A switch in series turns the whole circuit off. A fuse goes in series too.

A capacitor from a signal line to ground shorts AC noise to ground and leaves DC.

An integrated circuit is a tiny circuit in one package. You treat it as one part with numbered pins.

Gain means a bigger signal out than in. A transistor or an IC can provide it.

Gear and fixing problems

If it fails on every band, suspect the radio or the power. If it fails on one antenna, suspect that antenna.

WHAT A RECEIVER IS DOING

Sensitivity is hearing a weak signal. **Selectivity** is ignoring the ones next to it.
A mixer plus an oscillator make an intermediate frequency. That is a superheterodyne.
SSB is heard with a product detector. CW needs a beat oscillator so the tone is audible. FM uses a discriminator.

KNOBS THAT MATTER

- **Squelch** mutes the noise until a signal appears. Too high and you miss weak stations.
- **AGC** evens out loud and quiet signals.
- **Noise blanker** cuts pulse noise, like ignition.
- **Mic gain** too high makes a wide, distorted signal.
- Attenuator or “ATT” tames a too-strong signal.

INTERFERENCE

Harmonic: your transmitter also makes 2× and 3× the frequency. A low-pass filter at the transmitter reduces them. **Overload:** a strong station on its real frequency overwhelms a nearby receiver, like a TV or a phone. Fix that at the receiver: move the antenna, add a filter, or lower your power. Ferrite on the cables often stops RF getting into speakers and computers.

QUICK CHECKS

- No power: supply, fuse, power cord, battery.
- High SWR: coax, connector, antenna, water in the line.
- Repeater won’t answer: wrong offset, wrong tone, or you are on the output.
- Hear nothing: squelch, volume, tone, or the antenna is off.
- Distorted FM: mic gain, or you are off frequency.

DIGITAL HOOKUP

Most digital modes use a computer sound card or a small USB interface. The radio is on SSB for weak-signal modes and on FM for packet at VHF. Too much audio into the radio splashes across the band. Turn it down until the ALC just starts to move.

Modes and satellites

Use the narrowest mode that does the job. Bandwidth is how much spectrum you occupy.

Mode	About how wide	Remember
CW, Morse	150 Hz	On and off. Goes farthest for the power.
SSB voice	3 kHz	LSB below 10 MHz. USB above 10 MHz, including 10 meters.
AM voice	6 kHz	SSB is one side of an AM signal, and more efficient.
FM voice	10–15 kHz	The usual VHF/UHF mode. The stronger signal wins completely.
FT8, PSK31	very narrow	Weak-signal digital. Computer does the decoding.

DIGITAL YOU WILL SEE

- **Packet** digital messages, like a radio bulletin board.
- **APRS** position and short messages. 144.390 MHz in the US.
- **DMR, D-STAR, Fusion** digital voice. You need the right mode and often a talkgroup or room.
- **EchoLink and IRLP** tie radios together over the internet. Still identify. Still follow third-party rules.
- Slow-scan TV fits in a phone-width signal. Fast-scan TV is wide and lives on UHF.

SATELLITES

Most amateur satellites are low orbit. You can hear one for only a few minutes.

Uplink is you to the satellite. **Downlink** is the satellite to you. A **beacon** is its automatic ID.

Doppler: the frequency slides as the satellite moves. You retune. Use the published band plan. Don’t use a satellite uplink as a local chat channel.

TONES

CTCSS is a steady low tone that opens a repeater. You normally do not hear it. **DCS** is the digital version. **DTMF** is the keypad beeps, used to control a phone patch or a link. If the repeater is quiet and you are on the right frequency, the missing piece is usually the tone.

Antennas and feed lines

A half-wave dipole in feet is about 468 ÷ MHz. A quarter-wave vertical is about 234 ÷ MHz.

DIPOLE

Two equal wires, fed in the middle, about a half wavelength long. It radiates broadside, weakly off the ends. Hung flat, it is horizontally polarized.
Example: 468 / 146 ≈ 3.2 feet for a 2 meter dipole. Each side is half of that.

VERTICAL

A quarter-wave whip needs the other half: radials, or the metal body of a car. It radiates equally around the compass. Vertical polarization, which matches most FM repeaters.
A rubber duck is a short, lossy vertical. Fine up close. Poor at a distance.

YAGI, A BEAM

One driven element, a reflector behind it, directors in front. Gain goes forward. The sides and back are quiet. Use it to reach a far repeater, work a satellite, or find a hidden transmitter. More elements, tighter beam, more gain. Gain does not create power. It aims the power you already have.

FEED LINE

50 ohm coax is the normal cable. Loss rises with frequency, length, and cheap cable. Keep UHF runs short. PL-259 is the large HF plug. Handhelds use SMA or BNC. N connectors are better at UHF. Seal outdoor joints. Ladder line loses less but cannot be taped to a metal mast.

MATCH, DIRECTION, SAFETY

SWR tells you the match, not whether the antenna is a good radiator. A tuner at the radio protects the radio. Polarization should match the other station. To fox-hunt, use a beam or your body as a shield and walk toward the stronger signal. Never put a wire where it could fall into a power line.

Safety

Power lines, RF exposure, and charged capacitors cause the real injuries. Slow down around all three.

POWER LINES

Look up before you put up any mast. Assume every line is live. If an antenna or ladder could touch a line if it fell, it is too close. If a line is on your antenna, stay away and call the power company. Do not try to pull it off.

IN THE SHACK

Keep the third prong on the plug. It is the safety ground. Don’t replace a fuse with a bigger one. A capacitor in a power supply can shock you after the radio is unplugged. Discharge it. A shorted car battery or a ring across the terminals can burn you badly.

RF EXPOSURE

The things that raise exposure: more power, more duty cycle, more antenna gain, closer to people, and some frequencies more than others. Controlled means you and your household, who know the station is there. Uncontrolled means a neighbor. FM, digital, and RTTY are high duty cycle. SSB is lower because you are not transmitting full power the whole time. If the evaluation says you are over the limit, reduce power, move the antenna higher or farther, or transmit less. The rule is to evaluate, not to guess.

LIGHTNING AND GROUND

Disconnect outside antennas before a storm. Bond the radio, tuner, and antenna entry to a common ground, and bond that to the electrical service ground. A ground rod alone, not bonded, is not enough. Do not operate while a storm is on top of you.

TOWERS AND BATTERIES

Climbing means a harness, a hard hat, and a helper. Don’t climb alone. A lead-acid battery gives off hydrogen while charging. Ventilate it, and keep sparks away. A crank-up tower must be all the way down before anyone climbs it.

IF SOMEONE IS SHOCKED

Do not touch them while they are on the circuit. Turn the power off or knock the wire away with something dry and nonmetal. Then call for help. For an electrical fire, kill the power and use an extinguisher rated for electrical fires, not water.